

Assignment-4

Question-1:

Code:

```
// To find the sum of all integer numbers up to N where N is taken from the user.
#include <stdio.h>
int main() {
    int n, sum=0;
    printf("Enter the value of N: ");
    scanf("%d", &n);
    if(n<=0) printf("Please Enter a positive number.\n");
    else {
        for(int i = 0; i <n+1; i++) {
            sum+=i;
        }
        printf("The sum of all integer numbers from 1 to %d is %d.\n", n, sum);
    }
}
```

Output:

```
students@cse1ab1admin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_4$ ./q1
Enter the value of N: 10
The sum of all integer numbers from 1 to 10 is 55.
```

Question-2:

Code:

```
// To generate and print the first N elements of Fibonacci series.
#include <stdio.h>
int main() {
    int n, a=0, b=1, c;
    printf("Enter a number: ");
    scanf("%d", &n);
    if(n<=0) printf("Please Enter a positive number.\n");
    else {
        printf("The first %d elements of the Fibonacci series is: ", n);
        for(int i=0; i<n; i++) {
            printf("%d ", a);
            c=b;
            b=b+a;
            a=c;
        }
    }
}
```

Output:

```
students@cse1ab1admin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_4$ ./q2
Enter a number: 10
The first 10 elements of the Fibonacci series is: 0 1 1 2 3 5 8 13 21 34
```

Question-3:

Code:

```
// To add the first N terms of the exponential series for a given value of x
#include <stdio.h>
int main() {
    float n, x, res=0;
    printf("Enter the value of x: ");
    scanf("%f", &x);
    printf("Enter the value of N: ");
    scanf("%f", &n);
    for(int i=1; i<n; i++) {
        float fact=1, power=1;
        for(int j=1; j<i; j++) {
            fact *= j;
        }
        for(int k=1; k<i; k++) {
            power *= x;
        }
        res += (power/fact);
    }
    printf("The value of e^%.0f is %f\n", x, res);
}
```

Output:

```
students@cse-lab1-admin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_4$ ./q3
Enter the value of x: 1
Enter the value of N: 100
The value of e^1 is 2.718282
```

Question-4:

Code:

```
/* To print the multiplication table for a given number taken from
user. For example, if user tells table for 2 the table in the following
form must be displayed:
2x1=2
***
2x2=4
***
2x3=6
***
.
.
.
2x20=40
***
*/
#include <stdio.h>
int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    for(int i=1; i<=21; i++) {
        printf("%d * %d = %d\n", n, i, n*i);
    }
}
```

Output:

```
students@cse-lab1-admin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_4$ ./q4
Enter a number: 5
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50
5 * 11 = 55
5 * 12 = 60
5 * 13 = 65
5 * 14 = 70
5 * 15 = 75
5 * 16 = 80
5 * 17 = 85
5 * 18 = 90
5 * 19 = 95
5 * 20 = 100
```

Question-5a:

Code:

```
/* Pattern:
*
* *
* * *
* * * *
...
*/
#include <stdio.h>
int main() {
    int n;
    printf("Enter the number of rows: ");
    scanf("%d", &n);
    for(int i=1; i<=n; i++) {
        for(int j=1; j<=i; j++) {
            printf("* ");
        }
        printf("\n");
    }
}
```

Output:

```
students@cse1ab1admin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_4$ ./q5a
Enter the number of rows: 6
*
* *
* * *
* * * *
* * * * *
* * * * * *
```

Question-5b:

Code:

```
/* Pattern:
1
1 2
1 2 3
1 2 3 4
...
*/

#include <stdio.h>
int main() {
    int n;
    printf("Enter the number of rows: ");
    scanf("%d", &n);
    for(int i=1; i<=n; i++) {
        for(int j=1; j<=i; j++) {
            printf("%d ", j);
        }
        printf("\n");
    }
}
```

Output:

```
students@cse1ab1admin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_4$ ./q5b
Enter the number of rows: 6
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4 5 6
```

Question-5c:

Code:

```
/* Pattern:
1
2 3
4 5 6
7 8 9 10
...
*/

#include <stdio.h>
int main() {
    int n, a=1;
    printf("Enter the number of rows: ");
    scanf("%d", &n);
    for(int i=1; i<=n; i++) {
        for(int j=1; j<=i; j++) {
            printf("%d ", a);
            a++;
        }
        printf("\n");
    }
}
```

Output:

```
students@cse1ab1admin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_4$ ./q5c
Enter the number of rows: 6
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
16 17 18 19 20 21
```

Question-5d:

Code:

```
/* Pattern:
  1
 2 2
3 3 3
4 4 4 4
5 5 5 5 5
...
*/

#include <stdio.h>
int main() {
    int n;
    printf("Enter the number of rows: ");
    scanf("%d", &n);
    for(int i=1; i<=n; i++) {
        for(int j=n-i; j>0; j--) {
            printf(" ");
        }
        for(int k=1; k<=i; k++) {
            printf("%d ", i);
        }
        printf("\n");
    }
}
```

Output:

```
students@cslab1admin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_4$ ./q5d
Enter the number of rows: 6
  1
 2 2
3 3 3
4 4 4 4
5 5 5 5 5
6 6 6 6 6 6
```

Question-5e:

Code:

```
/* Pattern:
...
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
*/

#include <stdio.h>
int main() {
    int n;
    printf("Enter the number of rows: ");
    scanf("%d", &n);
    for(int i=1; i<=n; i++) {
        for(int j=1; j<=i; j++) {
            printf(" ");
        }
        for(int k=1; k<=n+2-i; k++) {
            printf("%d ", k);
        }
        printf("\n");
    }
}
```

Output:

```
students@cslab1admin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_4$ ./q5e
Enter the number of rows: 6
1 2 3 4 5 6
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```